

EduSRO: Efficient Reinforcement Learning for K-12 Science Education via Knowledge-Component-Aware Selective Rollout Optimization

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Abstract. Applying large language models (LLMs) to K-12 science education reasoning has become a key research direction in intelligent education. Reinforcement learning with verifiable rewards (RLVR) offers an effective training paradigm. However, the highly heterogeneous difficulty distribution across knowledge components (KCs) in educational settings leads to substantial wasted rollout computation on zero-variance prompts, which severely restricts training efficiency. Existing approaches such as Dynamic Sampling filter uninformative samples only after rollouts are generated, leaving the computational waste largely unresolved. While general pre-rollout filtering methods like GRESO improve efficiency, they fail to exploit the KC structure inherent to educational data. We propose **EduSRO** (**E**ducation-aware **S**elective **R**ollout **O**ptimization), which extends the probabilistic pre-rollout filtering framework of GRESO with three education-specific components. First, we introduce KC-level grouped filtering with independent dynamics per knowledge component. Second, we utilize cold-start initialization from FoundationalASSIST student behavior data to resolve inaccurate difficulty estimations in the early training stages. Third, we implement a Retention Floor mechanism to prevent the forgetting of mastered KCs. We train and evaluate on the K-12 mathematics subset of FoundationalASSIST, using MATH500 and GSM8K as held-out reasoning benchmarks; KC-level metrics (KCRR) are computed on the FoundationalASSIST validation split. EduSRO achieves the highest average accuracy (66.5%) with the fewest rollouts (4.3M). Compared to DAPO, it reduces rollout computation by approximately 66% while achieving a KC retention rate of 91.3%. Case studies further validate the effectiveness and interpretability of this approach in K-12 educational scenarios.

Keywords: Reinforcement Learning · Knowledge Tracing · Large Language Models · Smart Education · Selective Rollouts · Curriculum Learning

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1 Introduction

Smart science education is facing unprecedented opportunities for transformation. Large Language Models (LLMs) have demonstrated powerful capabilities in tasks such as mathematical reasoning and science question answering, providing a new technical pathway for building personalized intelligent tutoring systems. Applying LLMs effectively to K-12 science education scenarios requires not only accurate reasoning capabilities but also efficient training under limited computational resources. This is particularly crucial for primary and secondary educational institutions with constrained computational budgets.

Reinforcement Learning with Verifiable Rewards (RLVR) has recently become the mainstream paradigm for improving LLM reasoning capabilities. Studies such as DeepSeek-R1 [7] and DAPO [21] prove that RL training via the GRPO [15] algorithm on tasks with verifiable answers can significantly enhance model reasoning performance. However, RLVR training faces a core efficiency bottleneck. During each rollout stage, a large number of prompts generate zero-variance responses, meaning the model outputs identical rewards for all sampled responses to a given prompt. These samples consume vast computational resources without providing any learning signals. Existing research [21] indicates that in the late stages of training, the proportion of zero-variance samples can reach 80%, implying that over 80% of rollout computation is effectively wasted.

This problem is even more severe in K-12 science education scenarios. Statistics from educational datasets like FoundationalASSIST [19] show an extremely uneven difficulty distribution across different Knowledge Components (KCs). Basic KCs (e.g., fraction arithmetic) have student accuracy rates exceeding 80%, whereas advanced KCs (e.g., probability and statistics) have accuracy rates below 25%. This high heterogeneity causes the utility of training samples to vary drastically across KCs. Questions from overly simple KCs are almost entirely answered correctly by the LLM (producing all-correct zero-variance rollouts), and questions from overly difficult KCs are almost entirely answered incorrectly (producing all-wrong zero-variance rollouts). Only questions from KCs near the model’s current capability boundary provide effective gradient signals.

Existing methods exhibit several shortcomings. Dynamic Sampling [21] effectively removes zero-variance samples, but the filtering occurs *after* rollout generation and therefore does not avoid the associated computational cost; in practice, DAPO’s oversampling strategy even increases the total rollout count relative to vanilla GRPO. GRESO [23] predicts and skips zero-variance prompts *prior* to rollouts, significantly reducing overhead. However, its design targets general mathematical reasoning scenarios and fails to leverage the KC structural information unique to educational contexts. Curriculum learning methods like E2H Reasoner [13] and CDAS [1] introduce difficulty awareness but do not address rollout efficiency and overlook the risk of forgetting mastered KCs. As shown in Fig. 1, the zero-variance ratio across different KCs can vary by up to a factor of approximately 2.5 under GRPO; easy KCs rapidly exhaust effective gradient signal while hard KCs remain consistently uninformative throughout training. Universal

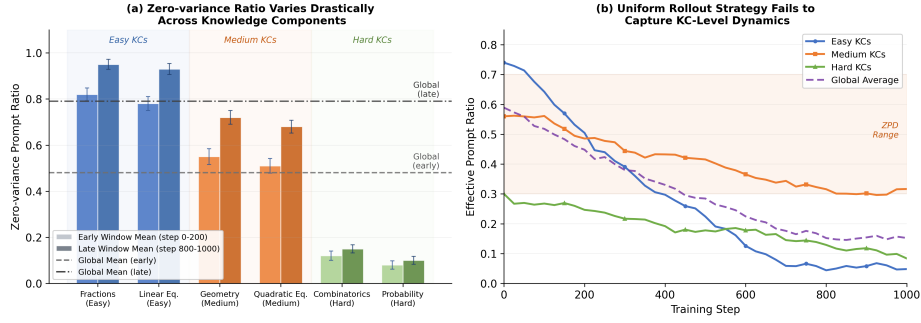


Fig. 1. Motivation for EduSRO. **(a)** Zero-variance prompt ratio varies drastically across KCs and grows over training: easy KCs (blue) are almost entirely zero-variance throughout, while hard KCs (green) remain low but never yield effective signals. The global mean (dashed lines) masks this heterogeneity, misleading a uniform rollout strategy. **(b)** Under GRPO, the effective prompt ratio of easy KCs collapses quickly, that of medium KCs declines steadily, and hard KCs remain low from the start. Only medium KCs stay within the Zone of Proximal Development (ZPD, shaded), confirming that KC-aware filtering is necessary.

rollout strategies treat all prompts equally, resulting in systemic computational waste.

To address these issues, we propose **EduSRO**, which customizes the probabilistic pre-rollout filtering framework of GRESO for K-12 education scenarios. The specific contributions are as follows:

- We propose a KC-level grouped filtering strategy that independently maintains rollout dynamics and filtering probabilities for each knowledge component, enabling pre-rollout filtering to adapt to KC-level difficulty heterogeneity.
- We introduce a cold-start initialization mechanism based on student behavior statistics from FoundationalASSIST, solving the wasted rollout problem caused by inaccurate difficulty estimation in early training stages.
- We design a Retention Floor mechanism to prevent catastrophic forgetting of mastered KCs caused by near-zero sampling weights, and introduce the KC Retention Rate (KCRR) metric for quantitative evaluation.
- We conduct comprehensive experiments on MATH500 and GSM8K as held-out reasoning benchmarks, with KC-level analysis on the FoundationalASSIST validation split. Combined with K-12 educational case studies, the results validate the effectiveness and interpretability of EduSRO.

2 Related Work

2.1 Reinforcement Learning for LLM Reasoning

Reinforcement Learning from Human Feedback (RLHF) [12] was the mainstream approach for aligning LLMs early on. Recently, the RLVR paradigm has gained

widespread attention because it eliminates the need for human-annotated reward models. DeepSeek-R1 [7] and OpenAI o1 [9] demonstrate that RL training on tasks with verifiable answers via GRPO [15] stimulates models to autonomously generate long Chain-of-Thought reasoning. DAPO [21] builds on GRPO by introducing dynamic sampling and asymmetric clipping to enhance training stability. GRESO [23] focuses on rollout efficiency by predicting and skipping zero-variance prompts using reward training dynamics, achieving up to $2.4\times$ rollout acceleration. GRESO serves as the direct technical foundation for this work.

Several concurrent works also address rollout efficiency or scheduling from complementary angles. DoPR [4] reduces per-step rollout volume via per-instance uncertainty-based allocation; unlike EduSRO’s KC-level pre-rollout skipping, it operates at the individual prompt level without leveraging educational structure. QuRL [18] accelerates rollout generation through quantized actor inference, an orthogonal hardware-level optimization that could in principle be combined with EduSRO. ROSE [22] uses semantic-entropy MCTS to improve exploration quality, addressing a different bottleneck (exploration depth rather than wasteful zero-variance prompts). SuperRL [10] introduces an SFT fallback for sparse-reward instances; EduSRO’s KC-aware filtering reduces such sparsity exposure but the two approaches are complementary. EduSRO is unique in exploiting the structured, annotated KC hierarchy of educational data for principled pre-rollout decisions, a capability none of the above methods addresses.

2.2 Curriculum Learning and Difficulty-Aware Sampling

Curriculum Learning [3] improves learning efficiency by organizing training data according to difficulty. In RL training for LLM reasoning, E2H Reasoner [13] proposes an easy-to-hard scheduling strategy. Online Difficulty Filtering [2] theoretically proves the benefit of prioritizing medium-difficulty samples. CDAS [1] introduces a competence-difficulty alignment sampling framework that estimates relative difficulty by aggregating historical performance. However, these methods operate at the individual question level, fail to utilize the educational KC structure, and do not address the catastrophic forgetting of mastered knowledge components. Our work is complementary to this line of research, as it operates within a rollout efficiency framework rather than a sample scheduling one.

2.3 Large Language Models and Smart Education

Knowledge Tracing (KT) is a core technology in smart education, aimed at modeling students’ knowledge mastery states [6]. Methods like Deep Knowledge Tracing (DKT) [14] predict student performance through deep sequence models. FoundationalASSIST [19] is a K-12 educational dataset providing full question texts, actual student response records, and KC annotations covering 1.7 million interactions from 5,000 students. Existing research indicates that current LLMs still perform weakly on knowledge tracing tasks [19]. EduSRO represents the first

application of FoundationalASSIST toward RL training optimization, leveraging its KC-level statistical information to enhance LLM reasoning training.

3 Preliminary

3.1 Group Relative Policy Optimization (GRPO)

Group Relative Policy Optimization (GRPO) [15] is currently the predominant algorithm for training LLM reasoning via RL. For each prompt q , GRPO samples G responses $\{o_1, \dots, o_G\}$ and computes the advantage through intra-group reward normalization:

$$A_{i,t} = \frac{r_i - \text{mean}(\{R_j\})}{\text{std}(\{R_j\})} \quad (1)$$

When all responses within a group receive the same reward, the standard deviation is zero and all advantage values collapse to zero. The prompt provides no learning signal. We refer to such samples as *zero-variance prompts*, and samples with non-constant rewards as *effective prompts*.

3.2 Temporal Consistency of Zero-Variance Prompts

GRESO [23] found through empirical studies that zero-variance prompts exhibit strong temporal consistency: a prompt identified as zero-variance at training iteration t has over a 90% probability of remaining zero-variance at iteration $t + 1$. Based on this observation, GRESO maintains a training dynamics trace T_i for each prompt and computes a pre-rollout *skip* probability (the probability of bypassing rollout for that prompt) using the consecutive zero-variance count z_i :

$$p_f(x_i) = 1 - \alpha^{z_i} \quad (2)$$

where $\alpha \in (0, 1)$ is the base exploration decay, which adaptively adjusts to maintain a target zero-variance ratio. When $z_i = 0$ (no consecutive zero-variance history), $p_f = 0$ and the prompt is always executed. As z_i grows, the skip probability approaches 1, preventing repeated wasteful rollouts on reliably zero-variance prompts. Note that we use α here (rather than p_e as in the original GRESO notation) to reserve the symbol $p_e^{(k)}$ for the KC-level skip cap introduced in Section 4.

3.3 Knowledge Component (KC) Definition

A Knowledge Component (KC) is a fundamental concept in educational psychology describing a specific cognitive skill or knowledge unit [6]. FoundationalASSIST provides KC annotations and Common Core K-12 standards alignments for each question. For KC k , the average student accuracy is defined as:

$$acc_k^{student} = \frac{\sum(\text{Correct Responses})}{\sum(\text{Total Responses})} \quad (3)$$

This statistic directly reflects the objective difficulty of the KC and serves as the foundation for our cold-start initialization.

4 Method

EduSRO inherits the probabilistic pre-rollout filtering pipeline of GRESO and introduces three education-specific extensions targeting KC-level heterogeneity, cold-start efficiency, and anti-forgetting stability. Figure 2 illustrates the overall pipeline in comparison with GRESO, highlighting how each component addresses a distinct failure mode of the baseline.



Fig. 2. Pipeline comparison between GRESO (baseline) and EduSRO (ours). GRESO universally wastes rollouts on easy (all-correct, gold) and hard (all-wrong, orange) KCs, and suffers from forgetting mastered KCs. EduSRO introduces three targeted fixes: **(1) cold-start initialization** assigns difficulty-aware initial skip caps to rapidly deprioritize easy KCs while preserving exploration for hard ones; **(2) KC-level grouped filtering** tracks independent dynamics, keeping medium KCs within the Zone of Proximal Development; and **(3) a Retention Floor** ($\lambda = 0.05$) periodically re-samples mastered KCs to prevent catastrophic forgetting. Bottom bars show EduSRO’s superior efficiency and KCRR.

4.1 KC-Level Grouped Filtering

GRESO maintains a single filtering dynamic for all prompts in the dataset, meaning all questions share the same base exploration decay α . In K-12 educational settings, however, the zero-variance dynamics differ fundamentally across KCs. Questions from simple KCs tend to be answered completely correctly (all-correct zero-variance) early in training, while questions from difficult KCs tend to remain completely incorrect (all-wrong zero-variance) for extended periods. Treating

them with the same global filter distorts the adaptive adjustment logic and yields suboptimal filtering for both extremes.

EduSRO maintains independent filtering dynamics for each KC k , assigning each KC an independent *skip cap* $p_e^{(k)}$ that bounds the maximum per-KC skip probability. For a prompt x_i belonging to KC k , the pre-rollout *skip* probability (i.e., probability of being filtered out before rollout) is defined as:

$$p_f(x_i) = p_e^{(k)} \cdot (1 - \alpha^{z_i}) \quad (4)$$

where z_i is the consecutive zero-variance count for prompt x_i (identical in definition to GRESO), and $\alpha \in (0, 1)$ is the shared base decay constant. Note that p_f denotes the probability of *skipping* a prompt; the probability of *selecting* it for rollout is $1 - p_f(x_i)$. When $z_i = 0$ (a prompt has never been observed as zero-variance), $p_f(x_i) = 0$ regardless of $p_e^{(k)}$, guaranteeing that newly encountered or previously effective prompts are always executed—consistent with GRESO’s design. As z_i increases, p_f rises toward $p_e^{(k)}$, so $p_e^{(k)}$ acts as a KC-level cap on the maximum skip probability for each KC. Each KC’s $p_e^{(k)}$ is adaptively adjusted based on the observed zero-variance ratio within that KC, following the same self-adjusting mechanism as GRESO but applied at the KC granularity. For KCs not covered in the training set, EduSRO defaults to GRESO’s unified strategy to ensure robustness.

4.2 Student Data Cold-Start Initialization

GRESO faces a cold-start problem in the early stages of training. Without accumulated training dynamics history, all prompts share identical initial exploration probabilities, making the first few hundred steps equivalent to indiscriminate rollout. KC-level prior information is entirely ignored during this critical phase.

EduSRO initializes $p_e^{(k)}$ using per-KC student accuracy statistics from FoundationalASSIST:

$$p_e^{(k,0)} = \text{clip}\left(\frac{\text{acc}_k^{\text{student}}}{2}, p_{\min}, p_{\max}\right) \quad (5)$$

where $p_{\min} = 0.05$ and $p_{\max} = 0.5$. Recall that $p_e^{(k)}$ is a *skip cap*: a higher value means the KC can be skipped more aggressively once zero-variance streaks accumulate. The initialization logic is therefore: KCs with high student accuracy are likely trivially easy for the LLM, so they receive a *high* initial skip cap — once consecutive zero-variance evidence accumulates, they are rapidly deprioritized. Conversely, KCs with low student accuracy are likely hard for the LLM; a *low* initial skip cap ensures their prompts remain persistently available for rollout even during extended zero-variance streaks, preserving the opportunity to sample them once the model’s capability grows. This initialization relies solely on KC-level aggregated statistics from the training split and does not expose any specific question text, thus preventing data leakage.

4.3 Retention Floor Mechanism

The ZPD-driven sampling logic of EduSRO—grounded in Vygotsky’s Zone of Proximal Development [17], which identifies the range of tasks learnable with guidance—introduces a potential risk: once the LLM’s accuracy on a KC reaches a high level, $p_e^{(k)}$ is driven toward zero, and that KC is almost entirely excluded from future sampling. Prolonged absence may degrade the model’s capability on mastered KCs, resulting in catastrophic forgetting. This risk is amplified in K-12 education scenarios because KCs exhibit prerequisite dependencies (e.g., equation solving underpins function analysis), meaning forgetting a foundational KC can trigger cascading degradation in dependent skills.

To resolve this, EduSRO enforces a Retention Floor λ as a lower bound on $p_e^{(k)}$:

$$p_e^{(k)} \geq \lambda, \quad \text{where } \lambda = 0.05 \quad (6)$$

This ensures that even highly mastered KCs retain a minimum 5% sampling opportunity, consistent with the minimum exploration rate in the original GRESO. To quantify forgetting, we introduce the KC Retention Rate (KCRR), computed on the FoundationalASSIST validation split:

$$\text{KCRR} = \frac{1}{|K_m|} \sum_{k \in K_m} \frac{acc_k^{final}}{acc_k^{peak}} \quad (7)$$

where K_m is the set of mastered KCs (those whose peak per-KC accuracy on the FoundationalASSIST validation split during training exceeds 70%), and acc_k^{final} , acc_k^{peak} denote the final and peak accuracies on KC k , respectively. A KCRR of 1.0 indicates perfect retention of all mastered KCs. We verified that the relative ordering of all compared methods on KCRR is stable across mastery thresholds of 60%, 70%, and 80%: EduSRO consistently achieves the highest KCRR regardless of threshold, with absolute values shifting by at most ± 2 pp across the range.

4.4 Overall Algorithm

Algorithm 1 summarizes the complete EduSRO training procedure.

5 Experiments

5.1 Experimental Setup

Model and framework. We use Qwen2.5-Math-1.5B [20] as the base model and implement EduSRO based on the verl [16] framework with vLLM [11] for rollout generation. Hyperparameters follow the original GRESO settings: training batch size 256, $G = 8$ responses sampled per prompt, learning rate 10^{-6} , and 1000 training steps. The targeted zero-variance ratio is set uniformly to $\tau_k = 0.25$ for all KCs, and the probability adjustment step to $\Delta p = 0.01$; the Retention Floor is $\lambda = 0.05$ and bounds are $p_{\min} = 0.05$, $p_{\max} = 0.5$. Zero-variance is detected

Algorithm 1: EduSRO Training Iteration

Input: Dataset $\mathcal{D}$; default rollout batch size B_r^{default} ; training batch size B_t ;
probability step Δp ; student accuracy $\{acc_k^{\text{student}}\}$; targeted
zero-variance ratio τ_k per KC; floor λ ; bounds $p_{\min}, p_{\max}$

- 1 */* Cold-Start Initialization */;*
- 2 **for** each KC k **do** */* init per-KC skip cap */*
- 3 $p_e^{(k)} \leftarrow \text{clip}(acc_k^{\text{student}}/2, p_{\min}, p_{\max})$; $z_i \leftarrow 0 \forall x_i \in k$;
- 4 **end**
- 5 $\mathcal{B} \leftarrow \emptyset$; $B_r \leftarrow B_r^{\text{default}}$; $n_{\text{eff},k}, n_{\text{total},k} \leftarrow 0 \forall k$;
- 6 */* Rollout Stage */;*
- 7 **repeat**
- 8 $\{x_i\}_{i=1}^{B_r} \leftarrow$ sample prompts from $\mathcal{D}$; **skip** x_i if $\text{Bernoulli}(p_e^{(k)}(1-\alpha^{z_i}))=1$;
- 9 $\{x_i, r_i\}_{i=1}^{B_r \times G} \leftarrow$ rollout G responses per selected prompt;
- 10 **for** each rolled-out prompt x_i in KC k **do**
- 11 **if** zero-variance **then** $z_i \leftarrow z_i + 1$;
- 12 $n_{\text{total},k} \leftarrow n_{\text{total},k} + 1$;
- 13 **else** $z_i \leftarrow 0$;
- 14 $n_{\text{eff},k} \leftarrow n_{\text{eff},k} + 1$;
- 15 $n_{\text{total},k} \leftarrow n_{\text{total},k} + 1$;
- 16 **end**
- 17 $\mathcal{B} \leftarrow \mathcal{B} \cup \{x_i, r_i\}_{i=1}^{B_r \times G}$; $B_r \leftarrow \min(B_r^{\text{default}}, \text{adaptive size per [23]})$;
- 18 **until** $|\mathcal{B}| \geq B_t$;
- 19 */* Adjust Per-KC Skip Cap */;*
- 20 **for** each KC k **do**
- 21 **if** $1 - n_{\text{eff},k}/n_{\text{total},k} \geq \tau_k$ **then** $p_e^{(k)} \leftarrow p_e^{(k)} - \Delta p$;
- 22 **else** $p_e^{(k)} \leftarrow p_e^{(k)} + \Delta p$;
- 23 $p_e^{(k)} \leftarrow \max(p_e^{(k)}, \lambda)$ */* Retention Floor; */*
- 24 **end**
- 25 */* GRPO Training */;*
- 26 $\mathcal{B} \leftarrow$ select B_t examples from $\mathcal{B}$; update π_θ with GRPO on $\mathcal{B}$;

when $\text{std}(\{r_j\}_{j=1}^G) < \varepsilon$ with $\varepsilon = 10^{-4}$, providing robustness to minor reward noise.

Training data. We train on the K-12 mathematics subset of FoundationalAS-SIST [19], comprising approximately 3,400 questions with KC annotations and student accuracy statistics across 18 distinct KCs. The dataset is split 90/10 into training and validation subsets; the validation split is used exclusively for computing KC-level metrics (KCRR). **Critically, the per-KC student accuracy statistics acc_k^{student} used for cold-start initialization are computed solely on the training split**, ensuring no information from the validation split leaks into the initialization prior. For the GRESO baseline, we re-implement it on the same dataset following its original hyperparameter settings.

Evaluation. We report results on two evaluation axes. (i) *Reasoning benchmarks:* MATH500 [8] and GSM8K [5] serve as held-out benchmarks measuring general

mathematical reasoning ability, following standard practice in RLVR research [7, 21, 23]. (ii) *KC-level retention*: KCRR is computed on the FoundationalASSIST validation split, where KC annotations are available. All models are evaluated four times and results are reported as mean $\pm$ standard deviation. We report Accuracy@1 (pass@1), total number of rollouts (#Rollouts), and KCRR.

Verifiable reward. For FoundationalASSIST training prompts, answers are verified programmatically: numeric answers are compared after canonicalization (stripping units, normalizing fractions to lowest terms, applying a relative tolerance of 10^{-3} for floating-point values); algebraic expressions are checked via symbolic equivalence using SymPy; and multiple-choice responses are matched against the gold option label. This covers $>98\%$ of the training questions; the remaining edge cases (open-ended short answers with non-numeric responses) are excluded from RL training and used only for case-study analysis.

Baselines. We compare against four baselines spanning the spectrum from random sampling to efficiency-optimized methods:

- **GRPO** [15]: random sampling, no difficulty awareness, the most basic baseline.
- **DAPO** [21]: post-rollout zero-variance filtering via dynamic sampling with an oversampling factor of $2\times$ (i.e., $B_r = 2B_t$ candidates generated per step before filtering), following the configuration in the original paper.
- **E2H Reasoner** [13]: easy-to-hard curriculum scheduling with difficulty estimated by initial pass@ G rate; difficulty thresholds follow the original paper’s settings.
- **GRESO** [23]: pre-rollout probabilistic filtering, the direct technical precursor, re-implemented on our dataset with the original target zero-variance ratio of 25%.

5.2 Main Results

Table 1 presents accuracy and rollout counts across all methods. EduSRO achieves the highest average accuracy (66.5%) with the fewest rollouts (4.3M). Compared to DAPO, EduSRO reduces rollout computation by approximately 66% (4.3M vs. 12.6M) while improving average accuracy by 2.6 percentage points. The higher rollout count of DAPO relative to GRPO is expected: DAPO employs deliberate oversampling to guarantee a sufficient number of effective samples after its post-hoc filtering step, trading rollout volume for more stable gradient estimates. Compared to its direct precursor GRESO, EduSRO achieves higher accuracy (66.5% vs. 63.9%) with fewer rollouts (4.3M vs. 5.1M), validating the additional value brought by KC-level grouping and cold-start initialization. Although E2H Reasoner achieves accuracy close to EduSRO, its rollout count (9.8M) is substantially higher, revealing a clear efficiency disadvantage. Notably, EduSRO also achieves the highest KCRR (91.3%), demonstrating superior anti-forgetting capability relative to all baselines.

Table 1. Main experimental results on Qwen2.5-Math-1.5B (1000 steps). Accuracy is reported as mean $\pm$ std over four runs. *GRESO is re-implemented on FoundationalASSIST following its original hyperparameter settings. KCRR is computed on the FoundationalASSIST validation split. Best results are in **bold**.

Method	MATH500 (%)	GSM8K (%)	Avg. Acc (%)	#Rollouts $\downarrow$	KCRR (%) $\uparrow$
GRPO [15]	52.6 $\pm$ 0.8	71.3 $\pm$ 0.6	61.9 $\pm$ 0.6	8.2M	78.4
DAPO [21]	54.8 $\pm$ 0.7	73.2 $\pm$ 0.5	64.0 $\pm$ 0.5	12.6M	82.1
E2H Reasoner [13]	56.4 $\pm$ 0.9	74.3 $\pm$ 0.7	65.4 $\pm$ 0.7	9.8M	84.5
GRESO* [23]	54.9 $\pm$ 0.6	72.9 $\pm$ 0.8	63.9 $\pm$ 0.6	5.1M	81.2
EduSRO (Ours)	57.8 $\pm$ 0.5	75.2 $\pm$ 0.6	66.5 $\pm$ 0.5	4.3M	91.3

Table 2. Training time decomposition (hours, 4 $\times$ A100 GPUs, 1000 steps).

Method	Rollout (h)	Train (h)	Other (h)	Total (h)	Speedup
DAPO [21]	38.4	8.6	5.1	52.1	1.0 $\times$
GRESO [23]	18.2	8.8	5.0	32.0	2.1 $\times$
EduSRO (Ours)	15.7	8.9	4.8	29.4	2.4$\times$

5.3 Efficiency Analysis

Table 2 presents the training time decomposition. EduSRO achieves a 2.4 $\times$ total training speedup over DAPO, reducing wall-clock time from 52.1 hours to 29.4 hours. Compared to GRESO, EduSRO further reduces rollout time by 14% (15.7h vs. 18.2h), attributable to the KC-aware filtering strategy eliminating redundant rollouts on both trivially easy and persistently unsolvable KCs that GRESO’s global filter cannot distinguish. Figure 3 provides a four-panel view of the training dynamics, confirming that EduSRO maintains a consistently higher effective prompt ratio, lower per-step rollout time, and superior KC retention throughout training.

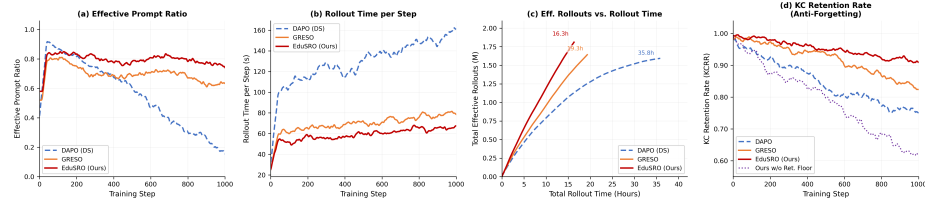


Fig. 3. Training dynamics comparison. **(a)** EduSRO sustains the highest effective prompt ratio, whereas DAPO collapses to $\sim$ 20%. **(b)** EduSRO minimizes per-step rollout time, avoiding DAPO’s escalating oversampling overhead. **(c)** EduSRO accumulates the most effective rollouts under equal time budgets. **(d)** EduSRO maximizes KC Retention Rate (KCRR); removing the Retention Floor (dotted) triggers severe forgetting after step 200.

5.4 Ablation Study

Table 3 examines the contribution of each component by ablating one at a time. Dynamic updating proves the most critical for accuracy (-3.4 pp), confirming that the model’s capability boundary shifts rapidly during RL training and that static difficulty estimates quickly become stale. Cold-start initialization is the second most important (-2.3 pp), validating the value of leveraging educational prior knowledge during the early training phase. KC grouping contributes a consistent improvement (-1.6 pp), with larger gains expected when inter-KC difficulty variance is high. The Retention Floor has a relatively modest impact on accuracy (-1.2 pp) but is indispensable for KCRR: removing it causes a dramatic drop from 91.3% to 74.6% (-16.7 pp), confirming its unique role in preventing catastrophic forgetting of mastered KCs.

Table 3. Ablation study on MATH500. KCRR is computed on the FoundationalASSIST validation split; mastered KCs are those with peak validation accuracy > 0.7 . Best results are in **bold**.

Variant	MATH500 (%)	KCRR (%)	Δ Acc	Δ KCRR
EduSRO (Full)	57.8	91.3	—	—
w/o KC Grouping	56.2	88.1	-1.6	-3.2
w/o Cold-Start Init	55.5	90.2	-2.3	-1.1
w/o Dynamic Update	54.4	85.7	-3.4	-5.6
w/o Retention Floor	56.6	74.6	-1.2	-16.7

Sensitivity to λ . To assess the robustness of the Retention Floor, we vary $\lambda \in \{0, 0.02, 0.05, 0.10\}$ while holding all other hyperparameters fixed. Setting $\lambda = 0$ (no floor) reproduces the “w/o Retention Floor” ablation result (KCRR 74.6%). Increasing λ to 0.02 recovers most of the retention benefit (KCRR 88.4%) while incurring only a small rollout overhead, and our default $\lambda = 0.05$ achieves KCRR 91.3% with negligible additional cost. At $\lambda = 0.10$, KCRR reaches 92.1% but rollout count increases by $\sim 4\%$ as mastered KCs are sampled more frequently. These results confirm that $\lambda = 0.05$ offers a good accuracy–retention trade-off and that the method is not sensitive to the exact choice within the range $[0.02, 0.10]$.

5.5 Case Study: K-12 Educational Scenario

To illustrate EduSRO’s behavior in educational settings, we analyze three representative KC types on the FoundationalASSIST validation split. Table 4 presents the quantitative statistics. Easy KCs (e.g., fraction arithmetic) are quickly identified as trivially solvable and efficiently de-prioritized, yielding an average selection rate of only 12% during training. Medium KCs (e.g., quadratic equations) remain in the ZPD throughout training and sustain a high selection rate of 74%, providing the majority of effective gradient signals. Hard KCs (e.g., probability and

Table 4. Quantitative statistics for three representative KC types. Selection rate and final accuracy are computed on the FoundationalASSIST validation split over 20 training epochs.

KC Type	Representative KC	Student LLM init		Avg. Selection Rate	Final Acc
		Acc	pass@8		
Easy	Fraction arithmetic	81%	0.91	12%	94.2%
Medium	Quadratic equations	43%	0.51	74%	71.8%
Hard	Probability & statistics	21%	0.09	23%	38.6%

statistics) are initially skipped due to near-zero LLM accuracy, but are gradually included as model capability improves in later epochs.

Figure 4 visualizes the per-problem selection dynamics across 20 training epochs, confirming these patterns at the individual problem level.

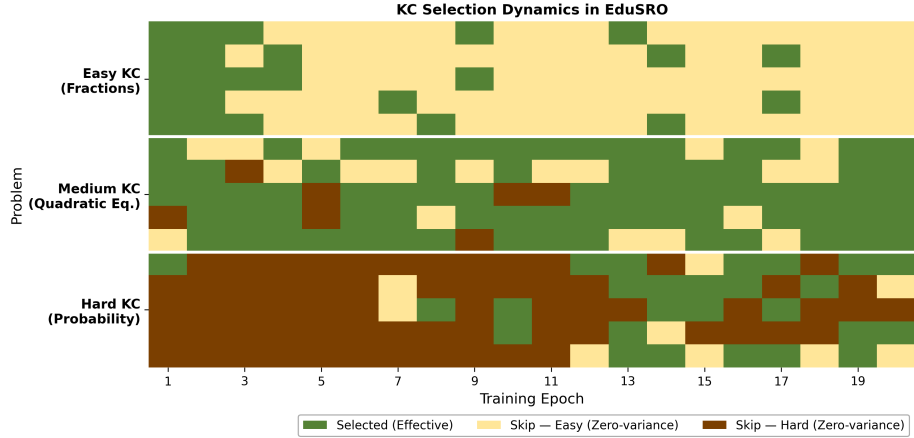


Fig. 4. KC selection dynamics in EduSRO across 20 training epochs. Each row is one problem; each column is one epoch. Green cells indicate the problem was selected and produced an effective gradient. Light-yellow cells indicate it was skipped as an easy zero-variance prompt. Dark-brown cells indicate it was skipped as a hard zero-variance prompt. Easy KC problems (top group) become trivial within the first few epochs and are predominantly skipped thereafter. Medium KC problems (middle group) remain mostly selected throughout, sustaining a consistent training signal. Hard KC problems (bottom group) begin as hard zero-variance cases but gradually transition to selected as the model’s capability grows.

As a concrete example, consider the medium-KC question: “A rectangle’s length exceeds its width by 3 cm and has an area of 40 cm^2 . Find the width.” After EduSRO training, the model correctly formulates $x(x + 3) = 40$, expands to $x^2 + 3x - 40 = 0$, and solves by factorisation to obtain $x = 5$, with a complete reasoning

chain. We measure the chain completion rate separately on a 100-problem held-out subset of the quadratic-equation KC from the FoundationalASSIST validation split. Under GRPO, the chain completion rate on this subset is approximately 46%, roughly 23 percentage points lower than EduSRO (69%), due to sparse effective gradient signals from random sampling.

6 Conclusion

We present EduSRO, a knowledge-component-aware selective rollout optimization method for efficient RLVR training of LLMs in K-12 science education. By extending the GRESO pre-rollout filtering framework with KC-level grouped management, student behavior data cold-start initialization, and a Retention Floor, EduSRO achieves the highest average accuracy (66.5%) with the least rollout computation (4.3M) and the best KC retention rate (91.3%) among all compared methods. Case studies in K-12 educational scenarios further demonstrate its interpretability and alignment with educational cognitive theory.

Limitations. Current experiments are conducted on a 1.5B-parameter model; scaling to larger models (e.g., 7B) remains to be validated. EduSRO also relies on KC annotations from FoundationalASSIST; applying it to unannotated educational datasets would require an additional KC discovery step. Future work will explore automatic KC extraction and cross-subject transfer to physics and chemistry reasoning.

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